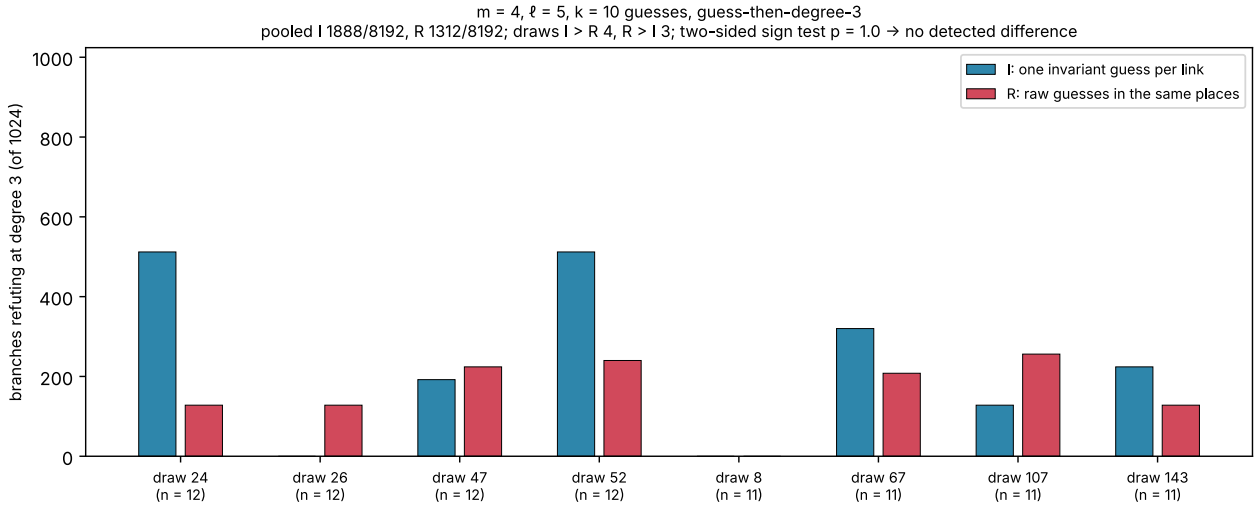
At m = 4, degree 3, one invariant guess per link is not detectably better than raw guesses

Study `research/dreg_m4_link_guess_d3_20261106/`, pre-registered. **Class:** stage diagnostic. **Registered verdict:** no detected difference. **Standing:** unverified assertion until a pcc verdict exists.

Question

At m = 3, invariant guesses on the chain's links did at least as well as raw guesses (2883, 3504). Every m = 4 cell this host can hold already refutes at degree 4 unguessed, so the question is asked at degree 3 with k = 10 guesses: does placing them on the three links' swap invariants (arm I) refute on more branches than raw guesses in the same places (arm R)?

Result



- Pooled over all 1024 branches of 8 draws: I refutes on 1,888 of 8,192, R on 1,312.
- By draw: I higher on 4, R on 3, both 0 on 1. The two-sided exact sign test gives p = 1.0, so the registered verdict is no detected difference.
- Controls valid: the instrument identity, the identity and exactness checks with arm I's link guesses re-derived from the basis, and every branch present.

Reading

- At m = 4 in the degree-3 regime the per-link invariant placement does not reliably help: two draws favour it strongly, two the other way.
- The m = 3 result stays a statement about m = 3. The degree-4 three-guess regime at m = 4 needs a host above 16 GB.
- No row moves, and rho is not reached.